

**Group Design Practical
Briefing Document Hilary Term 2026**

All second-year undergraduates reading Computer Science, Mathematics and Computer Science, or Computer Science and Philosophy take part in a group design exercise in their second year.

Goal

The goal of the group design practical is to give you, the student, a chance to understand and learn the skills required to succeed in delivering a real-world project, not just writing a program.

In addition to using the skills learnt during the four core programming courses to actually code, to succeed in this project you'll need to

- engage with your customer to understand what their needs are
- agree with the customer what your solution will and won't do in order to solve their needs (and update/negotiate changes with them as necessary)
- self-organise with your team to distribute work *and* make sure the whole solution works together
- plan your time to deliver what the customer needs when they need it
- provide the solution to the customer in a form
 - o they can easily deploy (installation, configuration)
 - o they can easily use (documentation, briefings, maintainability, good diagnostics)
 - o they won't break (careful testing).
 - o and considers the ethical and societal impacts of the solution.

These are all solid basics that any tech company or research group expects of any good contributor – this is your chance to develop these skills as part of your degree.

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Key Dates

Briefing Meeting	Wednesday 21 st January, 1pm	Week 1
OOP Lecture 1	Wednesday 21 st January, 2pm	Week 1
OOP Lecture 2	Wednesday 28 th January, 1pm	Week 2
OOP Lecture 3	Wednesday 28 th January, 2pm	Week 2
Deadline to fix team members	Monday 2 nd February, 12pm	Week 3
OOP Lecture - AI	Wednesday 4 th February, 1pm	Week 3
OOP Lecture 4	Wednesday 4 th February, 2pm	Week 3
Confirmation of project choices	Friday 6 th February, 12pm	Week 3
Briefing Meeting on specific projects		Week 4
Responsible Innovation Seminar	Wednesday 11 th February, 2pm	Week 4
Deadline to fix projects	Thursday 12 th February, 12pm	Week 4
Project Co-ordination and Management Seminar	Wednesday 18 th February, 2pm	Week 5
First progress meeting with external (and internal) supervisor – agree customer requirements and project specification and plan	Week 6 - Time and day to be confirmed by team and supervisor	Week 6
Individual mid-Term meeting with internal supervisor		Week 7
Software version control seminar	Wednesday 4 th March, 2pm	Week 7
Software testing seminar	Wednesday 11 th March, 2pm	Week 8
Second progress meeting with external (and internal) supervisor – Progress report, initial implementation and test results	Week 8 - Time and day to be confirmed by team and supervisor	Week 8
Third progress meeting with external (and internal) supervisor – delivery and final report	Trinity Term Week 2 - Time and day to be confirmed by team and supervisor	Week 2
Presentation event	Wednesday 13 th May	Week 3

Organisation

You will be allocated to a team of around 6 people before the Briefing Meeting. Swaps between teams are allowed until the deadline to fix team members – to confirm a swap both students involved should email student-projects@cs.ox.ac.uk.

Teams and projects will be allocated by a random selection process. Each team will be randomly paired with a project from the available list and assigned a supervisor accordingly. Once allocations are complete, teams will be notified of their assigned project and supervisor. Teams may subsequently swap projects with another team, provided that both teams and all individual team members involved agree to the swap. Further details about the project, can be found from page 6 of this document.

You will have an internal and an external supervisor.

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The external supervisor will act as a customer with whom the team engage to understand any detailed requirements for the project and to whom the team deliver their finished product.

The internal supervisor will act as a mentor, ready to help the team deliver their project.

They will arrange an individual mid-term meeting with each member of their team. This is so the student is able to discuss any concerns, the progress and contributions of individual students and the interaction with the industrial supervisor. The meeting should be arranged in week 7, once the first progress meeting has happened and before the second progress meeting. This meeting should be the default option and students may wish to opt out if they wish to do so.

It is the team's responsibility to drive the project, requesting help from the external and internal supervisor as necessary.

As part of the project, each team *must* do the following.

- Arrange an initial planning meeting, with their external (and internal) supervisor in Week 6 of Hilary Term. At this meeting the team should agree what they will be delivering, when, to the customer and agree with their mentor a plan for how they'll achieve that.
- Arrange a progress report meeting with their external (and internal) supervisor in Week 8 of Hilary Term. At this meeting, they'll present a progress report to the customer on their module implementations, test plans and test results. If the team and their internal and external supervisor agree, the team may instead provide a brief written report on their progress by the end of Hilary Term.

This meeting is also a good point to negotiate changes to the scope or timeline for what they are delivering (if for example it turns out that their project is significantly harder than originally expected, they can negotiate to reduce the scope). It is absolutely fine for teams to agree with the customer to change the scope of what they're delivering as long as the customer has reasonable warning (while it is absolutely not fine for example for the team to propose a reduction in scope 24 hours before the final delivery date!)

- Arrange a delivery meeting with their external (and internal) supervisor in Week 2 of Trinity Term. At this meeting, they deliver the agreed project to the customer, including the product, installation instructions, documentation, etc. as previously agreed.
- Arrange separate progress meetings with their internal supervisor unless the internal supervisor is able to attend progress meeting with the external supervisor. Each group should meet their internal supervisor at least three times.

In Week 2 of Trinity Term, each group will deliver to the internal supervisor a final project report and each student will also deliver to the internal supervisor a one-page summary of their individual contribution.

Teams are encouraged to engage with their external and internal supervisors more frequently than these three mandatory meetings.

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On Wednesday 13th May there will be an event to celebrate you and your hard work. Each group will be able to present their work and give a 5-minute talk, and there will be cash-prizes, too!

You'll receive detailed information nearer the time.

Overview and Marking

The group practical exercise is designed to take 20-30 hours of student time, mainly during Hilary Term (so a team of 6 delivers a project of 120-180 hours of work). The contribution of each student will be based on the group report, demonstration, and individual contribution sheet. This mark will count for approximately one third of the practical marks for the second year. Students must pass the group practical in order to pass the practical component of the course. The group design exercise will be marked on a scale of S-, S(pass), S, S+. These marks should be converted to a numerical mark using the following scale:

S+	100
S	60
S (pass)	40
S-	20

This exercise gives students experience of working against rigid deadlines, with a team of colleagues not of their own choosing, using externally supplied tools to undertake a fixed project. This will give some idea of the problems encountered in normal professional practice. Groups will be expected to exhibit professional skills in design, quality and management. Specifically, they will have to show that the work has been carefully planned, that components and systems have been properly tested, and that members of the group have cooperated effectively. The review meetings with mentors provide an opportunity to monitor group progress and for general discussion, but mentors will not be expected to provide technical advice or resolve technical issues. The responsibility for organising and completing the work lies squarely with the group members.

Our intention is that all group projects should be successful, and all students are expected to contribute to their own project in accordance with the timetable. Groups often encounter problems, and students are expected to manage these as they arise. However, if serious problems are encountered, such that members of the group are unable to resolve them, students should contact the academic admin team (academic.administrator@cs.ox.ac.uk).

Deliverables

Each part of the project gives rise to a set of deliverables *which must be given to the internal supervisor at the review meetings*. The internal supervisors will forward all group and personal reports to the examiners for inclusion in the practical work portfolio of all group members.

Specification and plan

Project topics are presented in the form of an outline design brief. Part of the work is to undertake a proper requirements analysis for the chosen project. However, it is important not to develop an over-elaborate specification which commits the group to more work than is necessary. The first major task is

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therefore to turn a relatively open and informal design brief into a more detailed project specification, setting out the major components of the system you will produce, what documentation you'll deliver, what diagnostics you'll provide, etc. Once this is agreed with your internal supervisor, you must produce a project plan which sets out who will do what, the time needed for developing and testing each module, dependencies between modules, etc.

When planning and executing the project, it is important to work to a budget. The entire project should take no more than 40 hours per team member, and records should be kept of time invested. Groups should set realistic targets and achieve them; there will be no additional credit for over-elaborate projects, or for individuals who offer or do more than is required. Each member of the group is expected to gain experience of programming in the course of the project – this may possibly involve test harnesses or scripts, data conversion utilities, a tutorial system, external interfaces, demonstration examples, installation packages, or other code and associated materials as appropriate to the project and the individuals in the team.

Module implementation and testing

The next task is to implement the components and test them. This is likely to require the construction of special test harnesses for separate classes. The deliverable for this task is a written *progress report* describing the testing procedures and results.

Project delivery and report

The final task is to piece the whole system together, test it and ensure that it is adequately documented. At the final meeting, the group must:

- Demonstrate the solution to the external supervisor.
- Deliver the solution and associated documentation in a format that the external supervisor can use.
- Submit a brief (approximately 4 pages long) group report to the internal supervisor describing the project aims, outcomes, successes and failures, and any lessons learnt. This report should include a short (approximately half a page) discussion of responsible innovation, exploring (i) the ethical issues you needed to consider during your design and development activities and (ii) the potential societal impacts (both positive and negative, intended and unintended) of the solution that you have developed.

Each team member must also submit a one-page summary of their own individual contribution to the internal supervisor.

Project A: Brain Barge

Squid Software

Inland shipping plays a vital role in container logistics between the seaports and inland terminals. It is the job of human planners to determine which barges visit which terminals, when, and which containers to load or discharge, while maximising transport capacity utilisation and on time performance. Currently, planners have access to all required data but lack intuitive visual cues and hard numbers to aid them in making the right decisions. They must manually combine, sort, and filter information about rotations, containers, vessel capacity, and terminal schedules in various software applications and Excel sheets to get an understanding of the puzzle they need to solve. The challenge is not only to visualise what's happening but to assist in making these decisions. At the end of the day, a planner needs to come up with an efficient plan while keeping all constraints given a set of barges, terminal visits, and containers that need to move within varying time windows. However, even a good plan suffers from reality. When conditions change, such as the closure of a major sea terminal, or a delay due to a broken lock, we need to mitigate the impact by making a new plan. By the end of the project, students will have created a prototype that allows planners to:

- Visually assess barge utilisation and routing choices.
- Understand the trade-offs between different planning scenarios.
- See and adapt to disruptions in near real-time.

The result will not only demonstrate technical and design ability but also show how data visualisation and intelligent algorithms can support complex logistical decision-making in practice.

Aims

- Develop a visualisation that gives clear insight into:
 1. The rotation of each barge (sequence of terminal calls over time).
 2. The loading and discharge per terminal (TEU/ton) and segment.
- Which containers are unassigned, delayed, or risk missing their closing times?
- (Bonus) Design an algorithm that automatically proposes a feasible rotation plan or allocation of containers to barges and terminals. The algorithm should explain its reasoning clearly, so that the planner can validate or adjust it.
- (Extra bonus) Demonstrate robustness to change. For example, simulate a disruption (e.g., a port terminal closes due to a strike causing an idling barge and a spike in capacity the next day) and show:
 1. How this is visualised in the interface.
 2. How your algorithm or interface supports re-planning or mitigation (alternative routes, other terminals, rescheduling).

Deliverables

- A web-based visualisation of barges, rotations, and container movements over time.
 1. A database / API consumes a provided dataset of:
 2. Barges and their properties (capacity, constraints).
 3. Rotations and historical calls from the past week.
 4. Containers to be transported next week (origin, destination, timing, priority).
- (Bonus) An algorithmic component that suggests rotation or assignment decisions, with human-interpretable justification.
- (Extra bonus) A change-handling demo that simulates disruptions and shows their effect on the plan and GUI.

Technology Requirements

- The web GUI preferably written in some Node.JS or TypeScript framework or vanilla JS. Most of Squid's current web interfaces are written in SvelteKit.
- If an API is developed any JS-related language / framework is fine.
- Algorithm may be written in Python, TypeScript, Rust, or Elixir and should run as a standalone service.

- The dataset will be supplied as JSON, XLSX, or a PostgreSQL dump.
- The project team will have access to a mentor familiar with real-world barge planning processes.

Project B: Intelligent hot desk management system Kuehne + Nagel

The hybrid working model becomes popular in the UK after the pandemic.

Many companies offer staff to work some days on-site and the remaining days from home in a week. For this reason, the concept of the 'Hot Desk' was introduced. The company does not need to provide a workspace that can accommodate all the employees, as not all of them work on-site on the same day. Instead, they offer fewer desks, which are shared by the staff. The employees have to book their desks for the day they work on-site. This working model can save the rental cost for the company, as well as the commute cost for the employee.

To ensure enough desks are available for the staff, it is essential to have a booking system for the employees to book the desks before their arrival. Some products are available in the market for this purpose. However, most of them only provide simple desk booking. They do not have the intelligence to assign desks to foster team interaction or capacity planning.

Our company will move to a new office applying the hot desk approach next year. We want to have a system not only provides basic desk booking but can also have the intelligence to foster team interaction, for example:

1. The same team members are automatically assigned to adjacent desks.
2. Related teams (e.g., Sales team, Marketing team) are automatically assigned to adjacent desks.

Moreover, it is helpful to capacity planning if the system can:

3. Learn the booking behaviour to predict the usage for a working day.

Objective

This project is to design and implement an Intelligent Hot Desk Management System. The platform can be web-based, mobile-based, or both. The project team can use their imagination to conceive a system fulfilling hot desk booking requirements, fostering team interaction, and capacity prediction. The above are just examples, and we are happy to see the project team propose different ideas.

Project C: Systemic Impact Explorer Using a Knowledge Graph Apex:E3

This project develops a knowledge-graph system that models how disruptions spread across a connected network (e.g., supply chains, transport, energy). Students will design a graph representing entities and their dependencies, ingest structured data, and implement propagation rules to simulate first-, second-, and third-order effects of an event.

The system outputs a clear "Systemic Impact Report" showing affected nodes, severity, and propagation paths, supported by visualisations. The project offers experience with graph modelling, multi-hop reasoning, and explainable impact analysis.

Project D: Multi-Agent Geopolitical Intelligence Assistant Apex:E3

This project builds an AI system where multiple specialised agents work together to monitor global news and generate concise geopolitical intelligence summaries. Using the Model Context Protocol (MCP), the system coordinates agents for news collection, event summarisation, and impact analysis. Students will integrate open-source models, retrieve data from sources such as GDELT or news APIs, and produce structured daily intelligence reports that highlight key actors, themes, and trends. The project provides hands-on experience with multi-agent orchestration, RAG workflows, and explainable analysis pipelines.

Project E: Private Route Intelligence: Differential Privacy for Competitive Graph Analytics in Logistics **Tigris**

In multi-tenant logistics platforms, competing companies could benefit massively from aggregate insights but cannot share sensitive operational data. For instance, in shipping logistics many companies have access to data which would be extremely beneficial to others in the industry. Shippers have accurate and timely data about ship movements which would allow freight forwards to better plan where and when they send cargo. Port authorities have extensive data about ship traffic and timings within ports which would greatly benefit shippers plan their port calls. However, the sensitive nature of this data deters companies from sharing their knowledge causing avoidable inefficiencies in the industry. A system which allows for aggregations or algorithmic outputs of private data to be shared without revealing the original raw data is sorely needed. The challenge combines cutting-edge privacy research with complex graph algorithms and modern ML techniques, requiring students to balance mathematical rigor with practical performance. This is an unsolved problem in industry with potential for real algorithmic innovation.

Aims

This project explores how differential privacy can unlock collaborative intelligence without exposing competitive secrets. Students will implement differential privacy mechanisms for sophisticated logistics operations ranging from graph analytics to predictive modelling. This includes privacy-preserving shortest path algorithms, differentially private graph neural networks for congestion prediction, private clustering to discover route patterns across competitors, and anomaly detection that doesn't reveal individual company behaviours. They'll explore how privacy guarantees degrade differently for one-shot queries versus iterative ML training, implement attacks to validate their privacy mechanisms, and potentially develop novel approaches that exploit logistics-specific graph properties for better privacy-utility trade-offs.

Deliverables

- Core Implementation: Choose and implement 3-4 operations from the following suggestions:
 - Private route optimization and capacity analysis queries
 - Graph neural networks for demand prediction with privacy guarantees
 - Clustering algorithms to find collaboration opportunities without revealing individual patterns
 - Anomaly detection on logistics flows while protecting tenant identity
 - Private shortest path algorithms with real-time constraints
 - Differentially private subgraph pattern mining
- Privacy Validation: Implementation of at least one attack (membership inference, model inversion, or reconstruction) against your chosen implementations
- Comparative Analysis: Evaluation showing privacy-utility trade-offs for your selected approaches
- Innovation Output: Novel algorithms, theoretical insights, or significant performance improvements (open-ended)

Technology Requirements

Python is recommended as the primary language, with libraries like NetworkX for graph operations, PyTorch/TensorFlow for machine learning, and NumPy/SciPy for numerical algorithms. Existing differential privacy libraries (OpenDP, Diffprivlib, Opacus for PyTorch) can serve as reference implementations or baselines. Students are free to use other languages where appropriate. We will provide logistics graph datasets with temporal features and multi-tenant structure. Students need only standard development machines. We offer mentorship on logistics domain expertise and real-world privacy requirements throughout the project.

Project F: Bringing ML sound classification to micro:bit CreateAI

Microbit

The Micro:bit Educational Foundation is a not-for-profit organisation that aims to inspire every child to create their best digital future. The BBC micro:bit is our microcontroller board focussed on education, with built-in sensors and an LED display. As of 2025 more than 11 million micro:bits have been distributed globally. In November 2024 the Micro:bit Educational Foundation launched [micro:bit CreateAI](#), a free, web-based tool that makes it easy for students (8-14) to explore AI through movement and machine learning. This project will work to improve the experience of students building and using these ML models. It potentially involves both embedded and web-based programming, and developing an understanding of machine learning techniques for constrained computation environments.

Aims

CreateAI today works with motion data. It is extremely effective for allowing students to follow an end-to-end ML training and deployment pipeline in under 20 minutes in a classroom setting. We'd like to see how we can bring the same easy end-to-end experience but with audio instead of motion data. This is interesting for students who are surrounded by audio-based AI devices like Alexa, Siri, etc. Therefore we'd like to add a prototype "audio" mode to the micro:bit CreateAI tool, that allows students to record audio samples, train models to classify samples, and ultimately deploy that model onto the micro:bit.

Deliverables

- We expect you to create and deploy a proof of concept branch of the [ml-trainer open source project](#) (from which micro:bit CreateAI is built) that works with audio, including deploying the model to the micro:bit.
- Updating the micro:bit CreateAI tool to capture audio samples, either via the micro:bit directly or from the computer's microphone. Probably it will be necessary to experiment with both to ensure good models
- Build a machine learning pipeline that runs in the browser that can create viable models to classify audio samples based on the training data. With CreateAI today we need just 3 samples of two different kinds of motion to be able to build a 'good enough' model. What do we need with audio?
- Using ml4f, or similar model compiler, deploy a model to the micro:bit that can classify samples in real time
- Documentation of the exploration and prototyping done

Technology Requirements

- This project requires you to work on a non-trivial open source project that uses TypeScript and React. We'll provide support to help orient the team but self-directed learning will also be needed.
- We will provide micro:bit devices for the team so they can fully experience the product and test running models on the device.
- We can assist with GitHub setup and deployment of demos.

Project: G: Micro:bit Digital Twin

Microbit

The Micro: bit Educational Foundation is a not-for-profit organisation that aims to inspire every child to create their best digital future. The BBC micro:bit is our microcontroller board focussed on education, with built-in sensors and an LED display. As of 2025 more than 11 million micro:bits have been distributed globally. In November 2024 the Micro:bit Educational Foundation launched [micro:bit CreateAI](#), a free, web-based tool that makes it easy for students (8-14) to explore AI through movement and machine learning. This project will work to improve the experience of

students building and using these ML models. It potentially involves both embedded and web-based programming, and developing an understanding of machine learning techniques for constrained computation environments.

Aims

Build a live, interactive digital twin of the BBC micro:bit that lives in the browser. It should be a visualisation of the device that is connected to a real, physical micro:bit and lets students interact with the real device and learn in the browser about the nature of the features of the micro:bit and how to program them. It also acts as a 'giant' version of the micro:bit for a teacher to project to the class. It works over a USB cable or Bluetooth.

For example, a user might connect the digital twin, and press a button on their physical micro:bit. The button in the digital twin will show as being pressed, and the screen will show example code that a user could use to change what the micro:bit does on button presses. Likewise, the user could shake the device and the digital twin will show a live view of the accelerometer data and display the code for the 'on shake' event. This is an explorative, research project. We want to be able to experiment with different ideas and there is a lot of scope for innovation and design.

This project is a new concept, and there isn't an existing codebase, so you will need to create a new Typescript/React project and setup deployment and testing pipelines (we can help you do this, in line with our other tech projects in the Foundation)

Use the micro:bit connection library to find, connect-to and provision the physical micro:bit

- Develop a strategy for how the digital twin and the physical micro:bit will communicate so that the twin can seamlessly display the behaviour of the real device
- Build a user interface for displaying the digital twin of the micro:bit and the 'hints' or 'documentation' accessible.
- If there's time or capacity in the team, make changes to the underlying C++ layer in micro:bit CODAL that instruments key API calls so that the digital twin can snoop events without needing a dedicated 'twin' program.

Technology Requirements

This project would be good for a team with web programming experience as well as embedded software (C/C++, but MicroPython would also be okay). UX and user interaction design will be useful skills. We will provide advice and/or an initial template to get you started with the web-based visualisation. We would typically use TypeScript and React for this sort of project but can adapt to the team's experience. We will provide micro:bit devices for the team for testing. We can assist with GitHub setup and deployment of demos.

Project: H: **Economics of Motherhood**

Purpose

To optimise motherhood through agency in the subject pillars of money, health and time via easy-to-use assessment and maintenance tools. Right now, the platform consists of an online course (almost finished) as a teaching tool, a website in development to feature other helpful ways to maximise motherhood. I will update when these become available.

Background

The Economics of Motherhood reframes motherhood as a system that can be assessed, managed and optimised with the same clarity as a business. It proposes the creation of AI or tech tools (VR?) that help mothers assess, track and improve their clarity and management of money, health and time; the three core resources which implicate quality of life and not least, productivity. By quantifying invisible work and any inefficiencies, and providing data driven feedback, the idea is to

Economics of Motherhood

provide a heightened sense of agency and measurable progress to help make these years as good as they could be, on several levels.

Possibilities

Designing and prototyping a digital platform – a dashboard or intelligent assistant that helps transform subjective experiences of motherhood into actionable insights, such as collecting data through quizzes or easily interactive tools to support self-management and personal growth. The way the programme parallels business is through regarding the mother as an asset, the importance of clarity in cash flow and spending, productivity (for example in assessing best time of day to do things based on daily energy level patterns), long-term future planning – practices that make business run well and grow. A mother is already CEO, CFO, COO, HR and more in a family. Technical areas can include, but not be limited to, AI and Data Science, UX/ Human Computer interaction, Systems and Integration, perhaps a ‘motherhood index’ that measures the three subject pillars into 1 wellbeing score – as a baseline to improve. And, of course, anything I haven’t thought of!

Possible Outcomes

This project could allow students to demonstrate technical skill, data empathy and social innovation. It might provide an opportunity for teamwork on a real-world concept with lasting human impact – giving mothers clarity over the unseen systems that shape their everyday lives.

Of Interest

Working mothers still take more responsibility at home, 60% average. To outsource a mother’s work would cost an average £120K per year. One of their biggest issues is long-term sleep deprivation. The length of time with a child is about 18 years, until the dynamic changes. My goal is to help mothers make the most of this treasured time within and beyond those years.

Project I:

Layla Smart Glasses

Compu-J

In live meetings, crucial follow-ups (e.g., budget, timeline, dependencies) are often missed because participants are juggling note-taking, recall, and conversation flow. The Even Realities G1 smart glasses provide a discreet display that can surface just-in-time cues without breaking eye contact. The goal is a real-time “copilot” that listens on-device during a meeting, detects short utterances, and gently prompts the wearer with glanceable cues like “clarify budget,” “ask timeline,” or “reference last quarter metrics,” drawing from a lightweight “company knowledge” corpus (agenda, past notes, one-pagers). The focus is one polished end-to-end demo that feels seamless, respects privacy (e.g., an explicit no-record mode), and helps a user make better decisions, faster.

Aims

- Develop a React web-based frontend, which can be compiled into a responsive native application.
 - This application should be able to record the microphone and system audio, and also the screen.
 - It should be able to stream the audio and visuals to an LLM for quick analysis.
- Develop a backend that can accept these audio and visual streams, and provide the data for the application.
 - Using a model router, we should be able to process and analyse incoming data.
 - High-speed inference to generate conversation suggestions and actions on the fly.
- Connect this system to a pair of Augmented Reality (AR) smart glasses (Even Realities G1) and overlay actions in the Heads-Up Display (HUD).
 - Display conversation actions and suggestions generated by the system in the overlay.
 - Consider taking live microphone input from the far-field microphone array in the glasses.

Deliverables

- A React based application providing controls, monitoring, and information about conversations.
- Python backend providing compute, databases, and application logic.
- Mobile solution to connect to a pair of smart glasses to overlay actions and suggestions.

- A robust LLM solution which responds to live user input with sub-1500ms P95.

Technology Requirements

1. The web frontend should be written with React and TypeScript/Javascript.
2. Backend is written with Python.
3. High-speed inference with modern LLMs.
4. Output to a provided pair of AR glasses with a HUD.
5. The project team will have access to a mentor familiar with production implementations.

Project J: **Gaming the Market** **OxFord Asset Management**

Financial markets are rich in patterns that humans can often recognise instinctively. Gamifying this intuition offers a novel way to explore whether people can make accurate short-term market predictions, especially in highly volatile environments such as cryptocurrency markets. This project explores the creation of a mobile game in which real-time market data defines the game world. Players pilot a spaceship through a cavern whose shape dynamically reflects short-term market movements via indicators such as Bollinger bands, with colours and effects conveying order-book depth and trading volumes. By navigating with momentum and limited reaction time, players implicitly forecast market evolution; their score corresponds to hypothetical trading performance.

Aims

- To design and implement a playable, mobile-friendly game that encodes short-term cryptocurrency price movements into an interactive environment.
- To investigate how well human gameplay aligns with profitable predictive behaviour.
- To build the supporting data pipeline and components needed to integrate real (or mock) market data into the game world.

Deliverables

- A working game prototype including:
 - o Title screen, gameplay loop, and end-game screen.
 - o User-controlled spaceship navigation with inertia and collision detection.
 - o Real-time rendering of the cavern, score, and relevant market-driven visuals.
- A data-processing module that converts a live or fixed cryptocurrency dataset into the cavern model.
- A simple networked client-server setup capable of streaming price/volume data (mock data acceptable).
- Initial tuning of cavern evolution and scoring to achieve good playability.
- (Optional stretch) An agent trained in a deep-learning framework (e.g., Gymnasium) that plays the game and can be benchmarked against humans

Technology Requirements

- Game engine: Unity (or similar) for graphics, input handling, and game-loop management.
- Programming languages: C#, Python, or others as appropriate for game logic and data pipeline.
- Market data feed: A live data source or a mock dataset; there are freely available live and historical datasets that could be used for development.
- Server framework: Lightweight client/server architecture to deliver streaming data.
- Machine learning environment (optional): Gymnasium or equivalent for reinforcement-learning agents.

Project K: **Project McIlroy: Beta+ Strategies 1** **Vanguard Asset Management**

This project offers an exciting opportunity to work at the intersection of data science, finance, and strategy. You'll help design a proof-of-value tool that empowers portfolio managers to make smarter

decisions during voluntary corporate events—such as syndicate deals, spin-offs, Dutch auctions, and rights issues.

- Motivation: Vanguard Equity portfolio managers manage more than \$7 Trillion USD globally & strive to track benchmarks closely, but market events such as spin-offs, Dutch auctions and rights issues create opportunities for outperformance.
- Overview: The project aims to develop a tool that uses historical back-testing to generate actionable signals for portfolio managers, helping them make decisions that maximize portfolio performance.
- Area: Investment management and quantitative analytics

Aims

- Create a model to scrape, categorise and interpret index methodology documents to infer key details like announcement dates, index implementation for corporate events & leveraging this to present expected outcomes per index provider.
- Analysis of historical price movements, index changes and market news around similar events to recommend optimal trading strategies.
- Summarisation of downside risks and portfolio impact as a % of risk budgets and exposure.
- Build a historical log of events to inform future decision-making by comparing predicted vs actual outcomes.

Deliverables

- POC (Proof of Concept) Application for portfolio manager use to determine decision making, including:
 - To participate / not participate in voluntary event
 - Signal confidence indicator
 - Maximum downside risk indicator (optional)

Technological Requirement

Include:

- Bloomberg Terminal

Project L: Driver Activity Logging and Management System (EU Transport Context) Samsara

In the European Union, drivers of vehicles above a certain weight are legally required to use a tachograph to record their working time and driving activities. A tachograph allows drivers to select from a very limited set of duty statuses, such as Driving, Rest/Break, Available, and Other Work. For drivers of lighter vehicles, a tachograph may not be required at all. While these systems ensure basic legal compliance, they do not provide organisations with a detailed view of what drivers are actually doing throughout their working day. As a result, transport and logistics companies often lack sufficient information to accurately calculate driver pay, analyse productivity, or understand how time is being spent. To address this gap, many organisations rely on additional manual processes, such as paper logs or spreadsheets, which are time-consuming, error-prone, and difficult to manage at scale. A digital system that allows drivers to record more detailed activities could greatly improve visibility into driver work patterns. For example, instead of only recording “Other Work”, drivers could log more specific activities such as loading, unloading, vehicle inspection, waiting at a depot, or administrative tasks. Providing tools to review, edit, and export this information would reduce the need for manual tracking and support more accurate reporting and payment.

Objective

The objective of this project is to design and implement a Driver Activity Logging and Management System. The system should allow drivers to record their daily activities in a simple and intuitive way, while enabling managers or administrators to review and manage this data.

The system should consist of:

- A mobile application that allows drivers to:
 - Select from a list of activities.
 - Start and stop activities during their working day.
 - View a history or timeline of their own activities.
 - Use the app without complex authentication (e.g. simple driver selection or log-on).
- A web-based dashboard that allows managers or administrators to:
 - View activity timelines for individual drivers.
 - Edit or delete recorded activities when corrections are needed.
 - Export driver activity data (for example, as a CSV file or via email) for use in payroll or reporting systems.
- A backend system that stores activity data and supports communication between the mobile app and the web dashboard.

The project team has significant freedom in defining detailed requirements and system behaviour. For example:

- Administrators could be allowed to configure custom activity types that drivers can select from.
- Visualisations could be added to summarise driver activity over time.
- Rules could be introduced to handle overlapping activities or corrections.

The focus of the project is on designing a practical, easy-to-use system that improves transparency of driver activities. The project offers a balance of mobile development, backend design, data modelling, and user interface work. Students are encouraged to be creative, propose additional features, and make design decisions that they feel best meet the needs of drivers and organisations.

Project M: AI-Powered PDF Sustainability Report Analysis

Ample

Background

- Brands often release lengthy sustainability or ESG reports, but extracting key information can be time-consuming. This project is aimed at accelerating the process of analysing these PDF reports by leveraging generative AI and advanced search capabilities.
- The goal is to create a web-based platform where PDFs of sustainability reports can be uploaded and then queried for instant retrieval of pertinent information. By enabling a “chat with the PDF” interface, users can ask natural-language questions (e.g., “Does Brand X hold XYZ accreditation?”) and receive answers pulled directly from the uploaded report.
- This area integrates document parsing, generative AI (LLMs), and sustainability reporting. The ability to quickly navigate and interrogate large PDF documents reduces friction in assessing brands’ ESG commitments.

Aims

- Develop a web-based system to upload sustainability PDF reports and convert them into searchable data.
- Implement an AI chatbot that can answer questions about the contents of each uploaded PDF.

Deliverables

- PDF Upload Portal: A secure user interface where companies/brands can upload their sustainability reports.
- Generative AI Chat Interface: The core feature that allows users to pose questions and receive context-aware answers based on the PDF’s content.
- Reporting/Highlighting Key Areas: A feature that highlights text passages in the PDF where the answers were found for improved transparency.

Project N: Interactive Home Heating Simulator: Heat Pump vs Gas/Oil/LPG/Direct Electric (Weather Driven Cost & Carbon) Heat Engineer

Homeowners increasingly need to decide whether to keep or replace fossil-fuel heating systems with heat pumps, but the decision is complex. Running cost, carbon impact, and performance depend on local weather, building heat-loss characteristics, heating system type, energy tariffs, and grid emissions factors. Many existing tools are either overly technical, not transparent about assumptions, or not location-aware. This project will build a homeowner-friendly interactive simulator with a graphical interface that compares a heat pump against gas, oil, LPG, and direct electric heating. It will use publicly available assumptions and real historical weather data for a selected region, producing clear and explainable outputs (annual cost, CO₂ emissions, energy use, and seasonal performance). The project is designed so that no NDA is required and no proprietary customer data is needed. Any proprietary Heat Engineer Software tools (if used) must be treated as black box inputs rather than implemented by the student team.

Aims

1. Build a web-based simulator that compares:
 - Air-source heat pump
 - Gas boiler
 - Oil boiler
 - LPG boiler
 - Direct electric heating
2. Use real weather data for a chosen location (historical or typical year profile) to drive heat demand and heat pump performance.
3. Allow homeowners to input basic building information and choose one of two heat-loss estimation modes:
4. Manual building archetypes (preset homes + insulation level)
5. Optional “Heat Engineer Whole-House Estimator” integration (black-box output input only)
6. Provide clear visual outputs: cost, carbon, seasonal performance, and sensitivity to key inputs (tariffs, insulation, flow temperature).
7. Document assumptions, limitations, and data sources transparently.

Key Features & Requirements

Inputs (User-facing)

- Location selection (with bundled example locations for offline demo reliability)
- Heat loss input mode selector:
 - A) Manual presets: typical buildings + insulation slider
 - B) Advanced option: use outputs from Heat Engineer’s Whole-House Estimator
 - Tariffs: electricity, gas, oil, LPG (global inputs; allow presets + user override)
 - Electricity grid emissions factor (global input; allow presets + user override)
 - Heating distribution / flow temperature preset (e.g., high-temp radiators vs low-temp UFH)

Simulation

- Heat demand estimated from outdoor temperature profile and heat-loss inputs
- Heat pump COP model based on outdoor temperature + flow temperature (generic performance curves or reference curves)
- Boiler models based on typical seasonal efficiencies
- Direct electric assumed ~100% conversion efficiency
- Cost and CO₂ computed over a year and summarised
- Outputs (Explainable + Visual)
- Annual cost comparison + annual CO₂ comparison across technologies

- Monthly breakdown charts
- COP vs outdoor temperature chart (heat pump)
- “Conditions where heat pump wins” visual
- Sensitivity controls (tariffs, insulation, flow temp)
- Optional report export (stretch)

Deliverables

1. A working prototype web application (runs locally; optional hosted deployment)
2. Weather-driven simulation engine + documented model assumptions
3. A set of bundled example weather datasets (5–10 locations) for reliable demonstration
4. A comparison dashboard for cost, CO₂, energy usage and performance metrics
5. Documentation: installation, usage, assumptions, limitations, and how to add new locations/tariffs
6. Final demo and short written report summarising approach, validation checks, and results

MVP & Stretch Goals

- One or more bundled location weather datasets
- Manual building pre-sets + insulation slider
- Comparison across heat pump, gas, oil, LPG, direct electric
- User inputs for tariffs and grid emissions
- Annual + monthly cost/CO₂ outputs with charts

Stretch Goals

- Global location search with API-based weather download + caching
- Country presets for tariffs and grid emissions (user override retained)
- Payback mode (installation cost assumptions + break-even years)
- Exportable homeowner report (PDF/HTML)

Technology requirements

- React / Next.js and charting library shadcn.
- TypeScript across frontend + backend
- Styling: TailwindCSS

Project O: **Trend-aware Risk Signals Engine for a Reputation & Exposure Manager Mycfo.ai (Squishy)**

Background

Individuals increasingly live with a long “digital footprint” of posts and comments that can resurface in high-stakes moments (e.g., job searches, promotions, public scrutiny). The meaning of content can also change as social and political context shifts—so something that felt harmless years ago may become sensitive when a related topic becomes prominent in the news. This project builds a transparent, explainable risk-scoring and UK trend-alert subsystem that can be integrated into a reputation and exposure manager. The subsystem will compute practical risk indicators for text items (posts/comments), generate clear reasons for flags, and monitor UK trending topics to raise alerts when historical items overlap with newly trending themes. The project sits at the intersection of online reputation management and trend monitoring, and is proposed by Squishy, an Oxford-based data-science start-up serving small businesses and entrepreneurs. This project is intended to produce a demonstrator and evaluation artefacts; the sponsor will develop the production scoring policy and commercial system independently.

Aims

- Build an explainable, transparent subsystem that helps individuals understand and manage their online reputation and exposure by surfacing potential risks in posts/comments with clear reasons. The project should include at least 5 “edge case” scenarios:

- o Examples: Sarcasm/irony, quoting someone else, ambiguous terms, political terms used in non-political contexts, a trend term with multiple meanings, etc.
- Model context shift (“suddenly risky”): demonstrate how the same historic item can move from low → medium/high risk when related topics become prominent in UK news, using a trend-recency factor and topic sensitivity multiplier.
- Ethical viability: outputs must be framed as risk indicators with explanations, not judgments about a person, and operate only on user-provided/consented data with clear limitations (for this project, a synthetic or public domain dataset will be provided by the sponsor).
- Explainability by design: for every risk flag, return the underlying signals and specific “reasons” (e.g., matched terms/categories, trend overlap) so users can make informed choices.

Deliverables

- Working codebase + setup: a single repo with clear README (install/run/test/demo), sensible structure, and configuration files for keywords/rules.
- Risk signals service (API): implement a risk-service that accepts an Item (post/comment text + timestamp + metadata) and returns a RiskScore with (a) a structured reason list and (b) score decomposition (per signal contribution). Baseline signals: sentiment, topic tags from config, age of post, and a rule-based toxicity indicator (no training required).
- Risk scoring design + backtesting: Propose a transparent scoring function (configurable weights/thresholds), justify design trade-offs (precision vs recall), and back test “suddenly risky” behaviour by re scoring historical items as UK trends change over time. Include (i) at least 3 case studies where an old item’s risk increases due to a trend spike, (ii) a brief error analysis (false positives/negatives) and how explanations reduce user harm, and (iii) at least 5 edge case scenarios (sarcasm/irony, quoting someone else, etc.).
- UK trending topics service: implement a trending-service that ingests a UK trend feed (e.g., UK headlines), extracts trend terms, and stores TrendTopic (term, volume, sentiment/tone, last seen) (or equivalent), with an endpoint to retrieve current trends.
- Trend cross match + alerts: implement a scheduled job that cross matches current UK trends to historical items (baseline: exact overlap) and creates an alert when overlap is found; optionally compute a simple “risk delta” (e.g., low → medium) using trend recency + topic sensitivity.
- Minimal demo dashboard: a lightweight UI showing (1) items/footprint list with risk buckets + filters, (2) “Why flagged?” explanation panel, (3) UK trend feed view, and (4) alerts view (“newly risky due to today’s trends”).
- Testing + short report: unit tests for core scoring/matching logic + basic API integration tests, add outcomes to the project write up.
- Optional stretch (only if ahead): similarity-based trend matching (e.g., TF IDF) instead of exact overlap.

Technology requirements

- Student-provided: laptops; Git/GitHub; Python 3.10+ (e.g., FastAPI); Node.js + TypeScript optional (if building a small API gateway); a local dev DB (SQLite or Postgres).

- Sponsor-provided (sponsor pack): synthetic/public-domain sample dataset (CSV/JSON) of Items; starter DB schema/migrations for Item, RiskScore, TrendTopic (and optionally Alert); starter keyword/topic list (YAML/JSON); API contract (OpenAPI YAML + example payloads); starter repo skeleton (.env.example, scripts).
- Out of scope: X/Instagram/Facebook/LinkedIn developer keys, OAuth apps— excluded for feasibility and compliance.
- External API access: a UK headlines/trends source may require an API key (e.g., NewsAPI). Sponsor will aim to support where needed and provide API keys.
- Only if chosen by teams: gated model tokens (e.g., certain hosted ML models). Sponsor will aim to support where needed and provide tokens.

Project P:

Risk Signal Prioritisation

EY

In many organisations, risk and compliance departments want to make use of data but sometimes find that data hard to interpret and take action on. The first step in their journey is often to create a set of risk signals that measure how often potentially problematic scenarios are happening (e.g., looking for multiple purchase orders with the same supplier as an indicator of employees attempting to circumvent purchasing approval limits). One challenge with this approach is that not all risk signals are equal - they are varied in terms of flagging rate (and false positive rate), severity of the scenario and volatility. The second challenge is that organisations rarely work the same way across the globe, using different business models and processes in response to local markets and regulations. The third, and most important, factor is time - risk and compliance departments do not have the resources to follow up on every potential issue and that is where you come in.

The goal of this project is to design a system that can prioritise specific areas for intervention based on a set of risk signals. Synthetic risk signal data will be provided covering a range of business activities and geographies. We anticipate you would prioritise specific business activities (and within those, specific risk signals) for each geography. We would suggest starting with anomaly detection approaches, but we are open to ideas. Depending on your interests, this project could steer towards:

- Sophistication / accuracy of the prioritisation process, perhaps incorporating other synthetic or public datasets
- Persuasive explanation (in words) of why your prioritised areas are important, particularly for readers who are less familiar with the underlying data
- A user interface to explore the risk signals and your prioritised areas, hopefully with insightful visuals
- Geographic prioritisation in addition to activity and risk signal prioritisation

As EY is an audit firm, students working on this project will be asked to disclose any organisations they might be viewed as controlling (e.g., as directors, trustees or significant shareholders).

Project Q: Project McIlroy: Beta+ Strategies 2

Vanguard Asset Management

This project offers an exciting opportunity to work at the intersection of data science, finance, and strategy. You'll help design a proof-of-value tool that empowers portfolio managers to make smarter decisions during voluntary corporate events—such as syndicate deals, spin-offs, Dutch auctions, and rights issues.

- Motivation: Vanguard Equity portfolio managers manage more than \$7 Trillion USD globally & strive to track benchmarks closely, but market events such as spin-offs, Dutch auctions and rights issues create opportunities for outperformance.
- Overview: The project aims to develop a tool that uses historical back-testing to generate actionable signals for portfolio managers, helping them make decisions that maximize portfolio performance.
- Area: Investment management and quantitative analytics

Aims

- Create a model to scrape, categorise and interpret index methodology documents to infer key details like announcement dates, index implementation for corporate events & leveraging this to present expected outcomes per index provider.
- Analysis of historical price movements, index changes and market news around similar events to recommend optimal trading strategies.
- Summarisation of downside risks and portfolio impact as a % of risk budgets and exposure.
- Build a historical log of events to inform future decision-making by comparing predicted vs actual outcomes.

Deliverables

- POC (Proof of Concept) Application for portfolio manager use to determine decision making, including:
 - To participate / not participate in voluntary event
 - Signal confidence indicator
 - Maximum downside risk indicator (optional)

Technological Requirement

Include:

- Bloomberg Terminal

Project R: Risk as Vibe

Core Technology

Background

The rapid emergence of “vibe coding” tools has fundamentally changed how software and digital capabilities can be created. These tools enable generalist professionals—such as management consultants without formal software engineering backgrounds—to quickly build interactive analytics, workflow tools, and even small full-stack or mobile applications. This shift is strategically attractive, as it dramatically lowers the barrier to innovation and accelerates time-to-value for clients.

However, this same acceleration introduces a critical gap in enterprise risk management. Capabilities created through vibe coding workflows typically bypass standard secure software development lifecycle (SDLC) pipelines, including code scanning, dependency analysis, vulnerability management, and policy enforcement. As a result, client-facing assets may unknowingly incorporate insecure libraries, outdated dependencies, or non-compliant architectural patterns, exposing firms and their clients to operational, security, legal, and reputational risk. This project addresses that gap by exploring how risk, security, and mitigation controls can be expressed and enforced “as code” directly within vibe coding workflows. Rather than forcing consultants to adopt heavyweight enterprise processes, the project aims to embed automated risk controls natively into the tools they

already use. The focus area sits at the intersection of secure software engineering, AI-assisted development, and enterprise risk governance, with the goal of enabling safe innovation at scale.

Aims

The primary aim of the project is to design and prototype a straight-through, automated “Risk as Code” capability that ensures security, compliance, and risk mitigation for software artifacts produced via vibe coding tools.

Specifically, the project aims to:

- Enable enterprise-grade risk and security controls to be applied to vibe-coded assets without disrupting creative workflows.
- Automatically detect, assess, and mitigate risks arising from insecure libraries, dependencies, configurations, or architectural patterns.
- Align vibe-coded development outputs with existing enterprise risk, security, and SDLC standards through policy-driven automation.
- Provide a scalable foundation that allows generalist consultants to safely build and deploy client-facing capabilities.

Deliverables

- A reference architecture for a “Risk as Code” framework embedded into vibe coding workflows.
- A working prototype or proof-of-concept demonstrating automated risk detection and mitigation within at least one major vibe coding tool.
- A library of codified risk, security, and compliance policies applicable to vibe-coded projects (e.g., dependency risk, data handling, authentication, deployment exposure)
- Integration patterns or adapters enabling use across common vibe coding tools such as Cursor, Claude Code, and Lovable.
- Documentation and usage guidance for consultants, including how risk controls operate transparently during development.
- A roadmap outlining how the capability could be scaled and integrated with the firm’s broader enterprise security and SDLC tooling.

Technology Requirements

The project will require access to and integration with modern AI-assisted development environments and supporting security technologies, including:

- APIs or extension frameworks for popular vibe coding tools (e.g., Cursor, Claude Code, Lovable) to enable embedded or inline risk checks.
- Dependency analysis and software composition analysis (SCA) tooling capable of programmatic invocation.
- Policy-as-code and rules-engine technologies to define and enforce risk and security standards automatically.
- Secure artifact inspection and static analysis capabilities tailored for lightweight, AI-generated codebases.
- Optional integration with existing enterprise security tooling (e.g., vulnerability databases, risk registers, approval workflows).

McKinsey can provide access to internal risk frameworks, security standards, and subject-matter expertise, as well as development environments for prototyping. External tooling or open-source components may be leveraged where appropriate to accelerate delivery and ensure interoperability with industry-standard vibe coding platforms.

Project S:	People/Support Assistant	Dojo
	Background	

Currently, the People team have six thousand tickets a year, most of which are simple queries that employees could easily do on their own but lack the knowledge to do so.

e.g.

- Where do I get my payslip?
- How do I get my P60, etc.)?

This wastes time for the People team and the employees could get this information much more quickly, than the current turnaround, if they knew where to look. In addition, we could analyse the queries and provide greater support, training and documentation for common queries.

Aims/ Deliverables

Develop an Assistant/chatbot with knowledge on the above topics that is integrated into Slack or Gemini to assist as a first port of call for People team queries, in order to reduce wasted time and increase the speed of request turnaround by the People Team.